

FARAHNAZ HOSSEINI, Ph.D.

Teaching Assistant Professor | Artificial Intelligence | Cybersecurity | Graph Machine Learning

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ACADEMIC PROFILE

Computer scientist and cybersecurity faculty researcher with expertise in trustworthy artificial intelligence, AI security, adversarial machine learning, graph neural networks, large-scale graph analytics, high-performance computing, and interdisciplinary AI applications. **Current Teaching Assistant Professor in the School of Cybersecurity at Old Dominion University.** Research spans secure and resilient graph learning, socioeconomic and public-health prediction, dynamic graph algorithms, biomedical AI, and responsible emerging technologies. Published researcher with **22+ peer-reviewed works, 60+ citations, and h-index 5**, with experience in student research mentoring, university teaching, and collaborative proposal development.

22+

Peer-reviewed works

60+

Citations

5

h-index

2026

Ph.D. in Computer Science

RESEARCH AREAS

Trustworthy AI & AI Security • Adversarial Machine Learning • Security & Privacy of AI Systems • Graph Neural Networks & Secure Graph Learning • AI for Cybersecurity • Large-Scale Graph Analytics & HPC • Multimodal / Generative AI Security • Responsible AI • Biomedical & Public-Health AI

ACADEMIC APPOINTMENTS

Teaching Assistant Professor | School of Cybersecurity, Old Dominion University | *Norfolk, Virginia*

Aug 2026 - Present

Teach undergraduate and graduate cybersecurity courses, including CYSE 201S: Cybersecurity and the Social Sciences, CYSE 602: Advanced Techniques for Cyber Defense, and CYSE 601: Advanced Cybersecurity Techniques and Operations.

Develop and deliver instruction spanning cyber defense, malware, network reconnaissance, security tools, social and ethical dimensions of cybersecurity, and emerging AI-security topics.

Mentor student research in AI-enabled cybersecurity, phishing detection, machine learning, natural language processing, cloud security, and data-driven security applications.

Build research directions at the intersection of trustworthy AI, graph learning, adversarial robustness, and cybersecurity while contributing to university and school service.

Graduate Teaching Assistant | University of North Texas | *Denton, Texas*

May 2025 - May 2026

Supported instruction in Social Issues in Computing and Data Visualization and guided project-based work using Python, Matplotlib, Plotly, Tableau, Power BI, and D3.js.

Provided grading, technical feedback, tutorials, and student support in undergraduate computing courses.

Graduate Research Assistant | University of North Texas | *Denton, Texas*

Jan 2022 - May 2025

Conducted NSF-funded research on scalable k-core and k-truss decomposition for large dynamic graphs using parallel and high-performance computing methods.

Developed graph neural network and machine-learning methods for socioeconomic predictive modeling with high-dimensional census and geospatial data.

Designed reproducible data, graph-construction, representation-learning, and predictive-modeling pipelines for large-scale experiments.

Computer Engineering Instructor and Lecturer | Payame Noor University

Jan 2015 - May 2015

Taught Artificial Intelligence and Research Methods; developed course materials, assessments, and student projects.

Supervised undergraduate projects and provided academic guidance.

Instructor | Technical and Professional Organization

Aug 2007 - 2020

Taught programming, web development, software engineering, databases, operating systems, networking/security, and software tools.

EDUCATION

Ph.D. in Computer Science | University of North Texas, Denton, Texas

May 2026

Dissertation: "Graph Neural Networks for Neighborhood Socioeconomic Modeling from High-Dimensional Census Data." Advisor: Dr. Ting Xiao

M.S. in Computer Engineering | University of Kashan

2014

Thesis: "Brain Tumor Detection and Recognition in MRI Images." Advisor: Dr. Hossein Ebrahimpour Komleh

B.S. in Computer Engineering | Kerman Higher Institute

2010

Associate Degree in Computer Hardware | Kharazmi Applied-Scientific Higher Education Institute - Tiran and Karvan Branch

2008

FEATURED RESEARCH & ONGOING PROJECTS

Trustworthy Graph AI Under Adversarial Attacks: Evaluating feature, structural, and mixed perturbations against GCN, GraphSAGE, and hybrid GNN-XGBoost models. Developing a Similarity-Consistency Filter (SCF) to suppress suspicious graph connections and improve model resilience under attack.

Graph-Based Public-Health Prediction: Extending geo-embedded census graph learning to predict diabetes, obesity, and frequent mental distress using XGBoost, GCN, GraphSAGE, and hybrid graph-machine-learning models.

Socioeconomic Predictive Modeling with Geo-Embedded Census Data: Developing graph-based and hybrid predictive models over U.S. Census and ZCTA-level geographic structure, including representation learning, KNN graph construction, community discovery, and interpretable socioeconomic modeling.

AI-Enabled Cybersecurity & Phishing Detection: Mentoring and developing student research that applies machine learning, NLP, and security analytics to phishing detection, cyber-defense workflows, and related AI-enabled security problems.

Scalable Dynamic Graph Analytics: Developing and evaluating k-core and k-truss methods for large evolving graphs using HPC, parallel algorithms, CUDA, OpenMP, and scalable computing techniques.

Biomedical & Medical Imaging AI: Research contributions in white-blood-cell detection, forensic age estimation, digital X-ray analysis, MRI skull stripping, and brain-tumor detection using computer vision and optimization methods.

SELECTED PUBLICATIONS & SCHOLARLY IMPACT

22+ peer-reviewed works | 60+ citations | h-index: 5 • Complete publication record available on [Google Scholar](#)

1. Hosseini, F., & Xiao, T. (2026). "Graph Neural Networks and Geo-Embedded Census Data for Socioeconomic Prediction: A Comprehensive Review." Academia AI and Applications Journal.
2. Hosseini, F., & Xiao, T. (2026). "Robust Socioeconomic Community Discovery from Census-Derived Indicators via KNN Graphs and Louvain Stability-Conductance Analysis." IEEE SoutheastCon 2026, Huntsville, Alabama.
3. Hosseini, F., & Xiao, T. (2025). "Discovering Socioeconomic Structure with Unsupervised Graph Neural Embeddings on Census Data." CSCI 2025, Las Vegas, Nevada.
4. Hosseini, F., & Xiao, T. (2025). "Graph-Augmented Machine Learning for Socioeconomic Prediction Using Census Data." CSCI 2025, Las Vegas, Nevada.
5. Hosseini, F., & Xiao, T. (2025). "Using Dynamic Graph Neural Networks for Temporal Analysis in Financial Markets." CMI 2025, Michigan.
6. Nickfarjam, A. M., Dalilian, M., Zargari, S. A., Farzandipour, M., Hosseini, F., & Xiao, T. (2025). "Comparison of Deep Learning Methods for Age Determination in Forensic Clients Aged 5–18 Years." CMI 2025.

7. Nickfarjam, A. M., Hosseini, F., Kia, Z. S., Shokrizadeh Arani, L., & Xiao, T. (2025). "White Blood Cell Detection and Counting Using PSO." *Studies in Health Technology and Informatics*, 328, 332–336. doi:10.3233/SHTI250731.
8. Hosseini, F., & Xiao, T. (2025). "Developing K-Core and K-Truss Algorithms in Large Dynamic Graphs." *SIAM Conference on Computational Science and Engineering (CSE25)*, Fort Worth, Texas.
9. Khanda, A., Shovan, S. M., Pandey, A., Hosseini, F., & Das, S. K. (2024). "Efficient Approaches to Analyzing Large Dynamic Networks." *SC24*, Atlanta, Georgia.
10. Biglari, S., Hosseini, F., Upadhyay, A., & Zhao, H. (2024). "Survey of Network-on-Chip (NoC) for Heterogeneous Multicore Systems." *IEEE MCSoc 2024*.
11. Bevara, R. V. K., Wagenvoort, I., Hosseini, F., Sharma, H., Nunna, V., & Xiao, T. (2024). "Census2Vec: Enhancing Socioeconomic Predictive Models with Geo-Embedded Data." *Intelligent Systems Conference (IntelliSys 2024)*, 626–640. Springer Nature.
12. Rafati, M., Kalantari, N., Azadbakht, J., Nickfarjam, A. M., & Hosseini, F. (2024). "Performance Evaluation of Fuzzy Genetic, Fuzzy Particle Swarm and Similar Insects' Optimization Algorithms on Denoising Problem Based on a Novel Combined Filter for Digital X-ray Images." *Multimedia Tools and Applications*, 83(6), 15483–15531.
13. Bevara, R. V. K., Xiao, T., & Hosseini, F. (2023). "Bias Analysis in Language Models Using an Association Test and Prompt Engineering." *IEEE QRS 2023*.
14. Rafati, M., Farnia, F., Romoozi, E., Nickfarjam, A. M., & Hosseini, F. (2021). "Measuring Panoramic Radiomorphometric Indices for Mandible Bone Using Active Shape Model and BIC-SVM." *International Journal of Imaging Systems and Technology*, 31(3), 1711–1730.

TEACHING, ADVISING & MENTORING

Old Dominion University: CYSE 201S - Cybersecurity and the Social Sciences; CYSE 602 - Advanced Techniques for Cyber Defense; CYSE 601 - Advanced Cybersecurity Techniques and Operations.

University of North Texas: teaching support for Social Issues in Computing and Data Visualization; project mentoring in Python and visual analytics.

Academic Mentor, Computing Research Association (CRA) UR2PhD Program: mentored undergraduate students in research methods, graduate-school preparation, parallel programming, CUDA, OpenMP, and graph algorithms.

Graduate Advisor, UNT AI Summer Research Program: advised graduate students on AI/data-science research, project planning, technical development, and academic writing.

Current student research mentoring includes AI-enabled cybersecurity, phishing detection, machine learning, NLP, cloud security, and data-driven security applications.

GRANT & PROPOSAL DEVELOPMENT

Contributed to research proposal development in artificial intelligence, machine learning, graph analytics, cybersecurity, data science, and interdisciplinary computing.

Prepared and refined technical narratives, literature reviews, research objectives, methodology, broader impacts, project plans, and supporting proposal materials.

Collaborated with faculty and research teams on research scoping and identification of internal seed-funding and external funding opportunities.

AWARDS, HONORS & GRANTS

Toulouse Graduate School Grant, University of North Texas, 2025.

Department of Computer Science and Engineering Grant, University of North Texas, 2025.

CSCE Grant, University of North Texas, 2025.

ACM SIGHPC Grant Recipient, Supercomputing 2024 (SC24), Atlanta, Georgia, 2024.

Student Volunteer, Supercomputing 2023 (SC23), Denver, Colorado; Supercomputing 2022 (SC22), Dallas, Texas.

Undergraduate Mentor, Computing Research Association (CRA) UR2PhD Program, 2023-2024.

Top Graduating Student, Department of Computer Engineering, University of Kashan.
Top 25% worldwide, KDD Cup 2013 “Author Disambiguation Challenge” (Track 2).
Certificate of Achievement, ACM International Collegiate Programming Contest (ICPC), Asia Tehran Regional.

PROFESSIONAL & UNIVERSITY SERVICE

Member, General Education Assessment Committee, Old Dominion University, 2026-2027.
Member, Hands-on Review Committee, Old Dominion University, 2026-Present.
Conference Reviewer, QRS 2026 - The 26th International Conference on Software Quality, Reliability, and Security.
Graduate Advisor, UNT AI Summer Research Program, 2023-2026.
Professional memberships: Association for Computing Machinery (ACM), IEEE, Computing Research Association (CRA), AnitaB.org, and Women in High Performance Computing (WHPC).

SELECTED PRESENTATIONS & POSTERS

Hosseini, F. (2025). “Developing k-core and k-truss algorithms in large dynamic graphs.” Oral presentation, SIAM CSE25, Fort Worth, Texas.
Pandey, A., Khanda, S., Shovan, F. M., Hosseini, F., Das, S., Norris, B., & Bhowmick, S. (2024). “Efficient approaches to analyzing large dynamic networks.” Poster, SC24.
Pandey, A., et al. (2023). “Scalable algorithms for analyzing large dynamic networks using CANDY.” Poster, SC23.

TECHNICAL EXPERTISE

AI / ML: Machine Learning, Deep Learning, Graph Neural Networks, GCN, GAT, GraphSAGE, Representation Learning, Predictive Modeling, NLP, Multimodal AI, Generative AI, LLMs

Trustworthy AI & Cybersecurity: Adversarial Machine Learning, AI Security, Secure Graph Learning, robustness evaluation, phishing detection, cyber defense, network reconnaissance, malware, security analytics

Programming & Data: Python, C++, SQL, NumPy, Pandas, scikit-learn, PyTorch, PyTorch Geometric, TensorFlow

Graph & HPC: Network Analysis, K-Core, K-Truss, KNN Graphs, Louvain Community Detection, OpenMP, MPI, CUDA, Parallel Algorithms

Visualization & Platforms: Matplotlib, Plotly, Tableau, Power BI, D3.js, Canvas LMS, Git/GitHub

CERTIFICATES & PROFESSIONAL DEVELOPMENT

Google Cybersecurity Professional Certificate; Google AI Essentials Professional Certificate.
Duke University coursework/certificates in LLMops, Azure, AWS for AI, GenAI & LLMs, Advanced Data Engineering, and Databricks to Local LLMs.